

ENME 302 – Test 1 – 2025

Name:

Student Number:

Please answer **ALL** questions.

This test is worth 25% of your overall class grade.
Each problem statement has the associated mark value stated.

Do not turn over your cover sheet to start until you are told to begin. Extra blank pages are given for each problem with what should be enough room to answer each question, especially if you use both sides. However, if you need extra paper, please just ask and it will be provided.

If you separate the pages, ensure that they are stapled together again when handed in.

There is a total of 42 marks (20 marks for Question 1 and 22 marks for Question 2)

90 minutes total time allowed (plus an additional 5-10 minutes to upload your code).

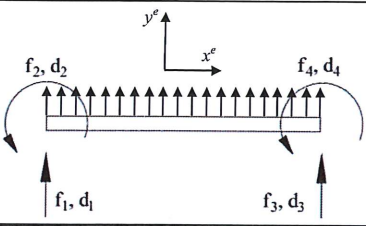
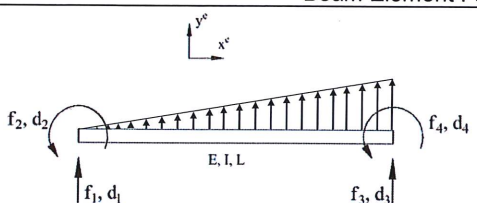
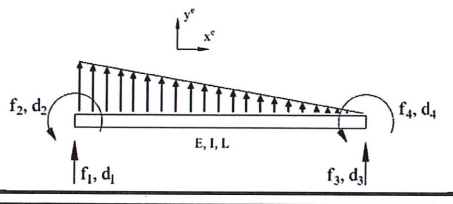
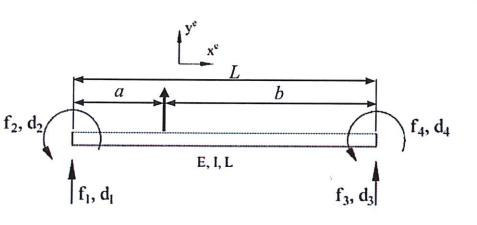
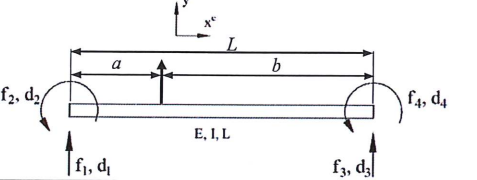
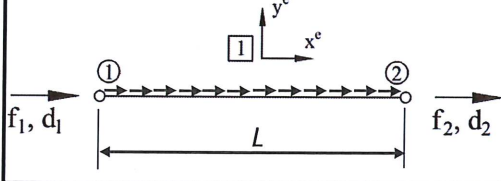
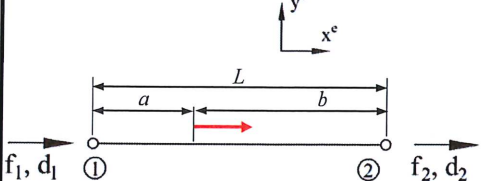
Circle or box all final answers! You may use pencil but make sure to highlight and/or ink final answers.

Label each part of each question, include appropriate units, and an appropriate number of significant figures, for all answers.

Some equations for equivalent nodal loading are presented on the next page for your reference.

At the completion of the test, you are asked to upload the code that you used to complete the test via the link on the course Ako | Learn page. While there are no marks allocated for your code, having access to your code helps with the awarding of partial credit when final answers are incorrect.

The equivalent nodal loading definitions (in element co-ordinates) are provided for reference.

Equivalent nodal loading definition for a Uniformly Distributed Load (UDL)		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,UDL}^e = \begin{Bmatrix} \bar{w}L/2 \\ \bar{w}L^2/12 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$	$f_{eq,UDL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$
Equivalent nodal loading definition for a Linearly Varying Load (LVL)		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,LVL}^e = \begin{Bmatrix} 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$	$f_{eq,LVL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 0 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$
	$f_{eq,LVL}^e = \begin{Bmatrix} 7\bar{w}L/20 \\ \bar{w}L^2/20 \\ 3\bar{w}L/20 \\ -\bar{w}L^2/30 \end{Bmatrix}$	$f_{eq,LVL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ 7\bar{w}L/20 \\ \bar{w}L^2/20 \\ 0 \\ 3\bar{w}L/20 \\ -\bar{w}L^2/30 \end{Bmatrix}$
Equivalent nodal loading definition for a transverse point load		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 1 - \frac{3a^2}{L^2} + \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{2a^2}{L} + a \\ \frac{3a^2}{L^2} - \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{a^2}{L} \end{Bmatrix}$	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 0 \\ 1 - 3(a^2/L^2) + 2(a^3/L^3) \\ (a^3/L^2) - 2(a^2/L) + a \\ 0 \\ 3(a^2/L^2) - 2(a^3/L^3) \\ (a^3/L^2) - (a^2/L) \end{Bmatrix}$
Specific, equivalent nodal loading definition for a mid-span point load, where $a = L/2$		
	$f_{eq,PL}^e = \begin{Bmatrix} \bar{w}/2 \\ \bar{w}L/8 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$	$f_{eq,PL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}/2 \\ \bar{w}L/8 \\ 0 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$
Equivalent nodal loading definition for a Distributed Axial Load		
Bar Element Formulation		Frame Element Formulation
	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ L/2 \end{Bmatrix}$	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ 0 \\ 0 \\ L/2 \\ 0 \\ 0 \end{Bmatrix}$
Equivalent nodal loading definition for a Concentrated Axial Load		
Bar Element Formulation		Frame Element Formulation
	$f_{eq,axialPL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ a/L \end{Bmatrix}$	$f_{eq,axialPL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ 0 \\ 0 \\ a/L \\ 0 \\ 0 \end{Bmatrix}$

Problem 1:

Blast deflection walls are common at airports and are used to deflect the hot, high-speed air ejected from an aircraft jet engine to protect buildings, vehicle traffic and ground personnel. A deflection structure is modelled in 2D as two frame elements, as shown on the next page.

The two boundary conditions at the base are considered to be fixed supports. A horizontal distributed pressure of intensity 0.8kPa is applied to the left of the structure to represent the pressure from the jet engine exhaust. Assume that the structural frame supports a length (into the page) of 5m of blast deflection wall.

Assume that both elements are frame elements, made of steel, and that both elements have the same length, cross-sectional area and second moment of area, with values given with the sketch on the next page.

[20 Total Marks]

- a. Draw a sketch of the structure showing the global degrees of freedom q for the model. **[2 Marks]**
- b. Draw a free-body diagram for each element, showing the element degrees of freedom in global co-ordinates as well as the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- c. Write out the assembly matrices A^e for each element **[2 Marks]**

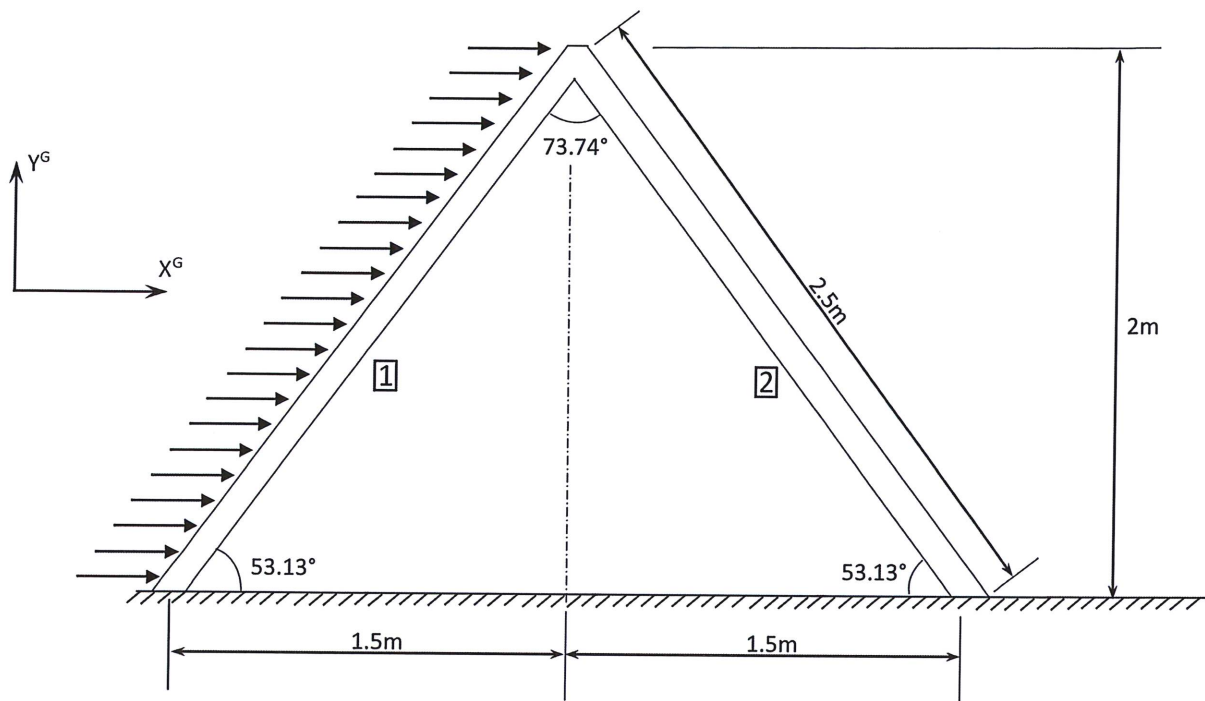
- d. Assume that the frame being modelled supports a section of wall (the distance into the page) of 5m, determine the loading on element 1 (in kN/m) from the given value of wind pressure of 0.8kPa.

Write out the element equivalent forcing terms for element 1 (f_{eq}^1) to represent the wind loading. Present both the axial and shear components of the distributed load (these will be a 6x1 vectors), as well as the final overall global forcing vector, Q .

If you cannot calculate the intensity of the uniformly distributed load that represents the roof loading, use a magnitude of $\bar{w}$ of 6 kN/m.

Note: The formulae on the second page of this test may be a useful reference. If you are unsure how to break up the shear and axial components of the distributed load, just apply the wind load as a pure shear load and continue with the question. **[4 Marks]**

- e. Solve for the overall deflections of the system, q , and present your result. **[1 Mark]**
- f. Sketch (by hand) the original and deflected shape of the structure, indicating the direction and magnitude of all deflection components (displacements and rotations). The sketch does not need to be to scale, but each component of deflection must be labelled. **[3 Marks]**
- g. Provide the reaction components (forces and moments) at all supports, parallel and perpendicular to the supports. Provide a sketch of these reaction loads, showing the magnitude and direction of all reaction components. **[3 Marks]**
- h. Determine the maximum bending moment magnitude (positive or negative) induced within element 2 (M_{max}^2) and state where it occurs within the element. **[1 Mark]**
- i. Determine the axial normal strain in element 2 (from axial loads only, ignoring any bending effects), and indicate whether it is tensile or compressive. **[1 Mark]**
- j. If the maximum magnitude of the axial normal strain in element 2 (from axial loads only, ignoring any bending effects), must not exceed $300\mu\epsilon$, what is the maximum permissible blast/wind load pressure that can be applied to the structure (in kPa)? **[1 Mark]**



All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = 200 \text{ GPa}$

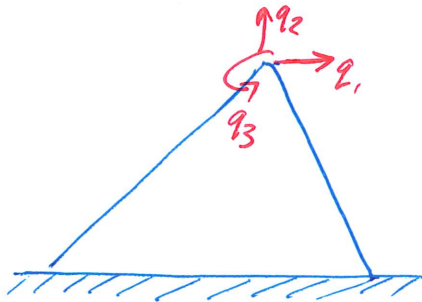
Second Moment of Area: $I^1 = I^2 = 3 \times 10^{-6} \text{ m}^4$

Cross-sectional Area: $A^1 = A^2 = 1 \times 10^{-4} \text{ m}^2$

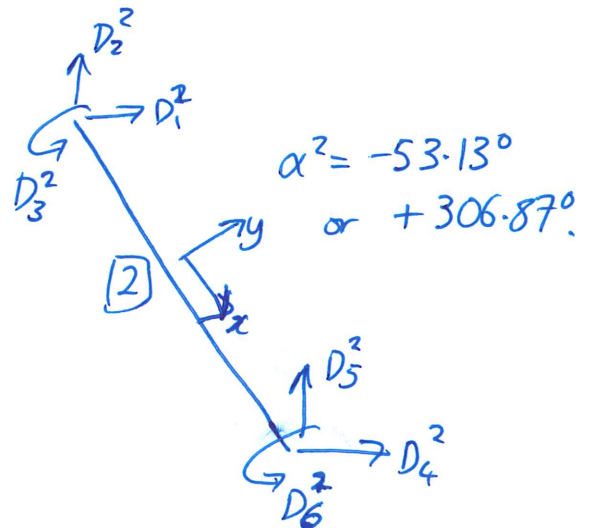
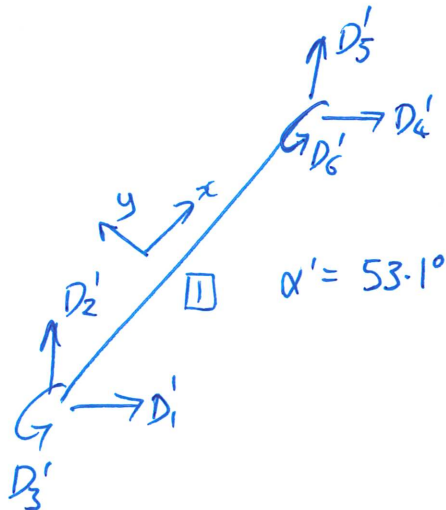
Lengths: $L^1 = 2.5 \text{ m}$

$L^2 = 2.5 \text{ m}$

a)

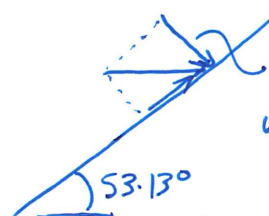


b)



$$c) \quad A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} \quad A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$d) \quad \text{Intensity} = \text{pressure} \times \text{depth} \\ = (0.8 \text{ kN/m}^2)(5 \text{ m}) \\ = 4 \text{ kN/m.}$$



$$\bar{w} = -\text{Intensity} \sin(53.13^\circ) \\ \bar{w} = -3,200 \text{ N/m}$$

$$\bar{p} = +\text{Intensity} \cos(53.13^\circ) \\ \bar{p} = +2,400 \text{ N/m.}$$

$$f_{eq,UDL}^1 = \begin{Bmatrix} 0 \\ \bar{w}L^1/2 \\ \bar{w}(L^1)^2/12 \\ 0 \\ \bar{w}L^1/2 \\ -\bar{w}(L^1)^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \\ (-3,200)(2.5)/2 \\ (-3,200)(2.5)^2/12 \\ 0 \\ (-3,200)(2.5)/2 \\ -(-3,200)(2.5)^2/12 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0 \text{ N} \\ -4,000 \text{ N} \\ -1,666.6 \text{ Nm} \\ 0 \text{ N} \\ -4,000 \text{ N} \\ +1,666.6 \text{ Nm} \end{Bmatrix}$$

Note: $\bar{w}$ is negative as the distributed shear load acts in the negative y^e direction. $\bar{p}$ is positive as the distributed axial load acts in the positive x^e direction for this element.

$$F_{eq,UDL}^1 = (\Delta^1)^T (f_{eq,UDL}^1) = \begin{Bmatrix} 3,200 \text{ N} \\ -2,400 \text{ N} \\ -1,666.6 \text{ Nm} \\ 3,200 \text{ N} \\ -2,400 \text{ N} \\ +1,666.6 \text{ Nm} \end{Bmatrix}$$

$$Q_{eq,UDL}^1 = A^1 F_{eq,UDL}^1 = \begin{Bmatrix} 3,200 \text{ N} \\ -2,400 \text{ N} \\ +1,666.6 \text{ Nm} \end{Bmatrix}$$

d) - continued.

$$f_{eq, DAL}' = \begin{Bmatrix} \bar{p}L'/2 \\ 0 \\ 0 \\ \bar{p}L'/2 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} (2,400)(2.5)/(2) \\ 0 \\ 0 \\ (2,400)(2.5)/(2) \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 3,000 N \\ 0 N \\ 0 Nm \\ 3,000 N \\ 0 N \\ 0 Nm \end{Bmatrix}$$

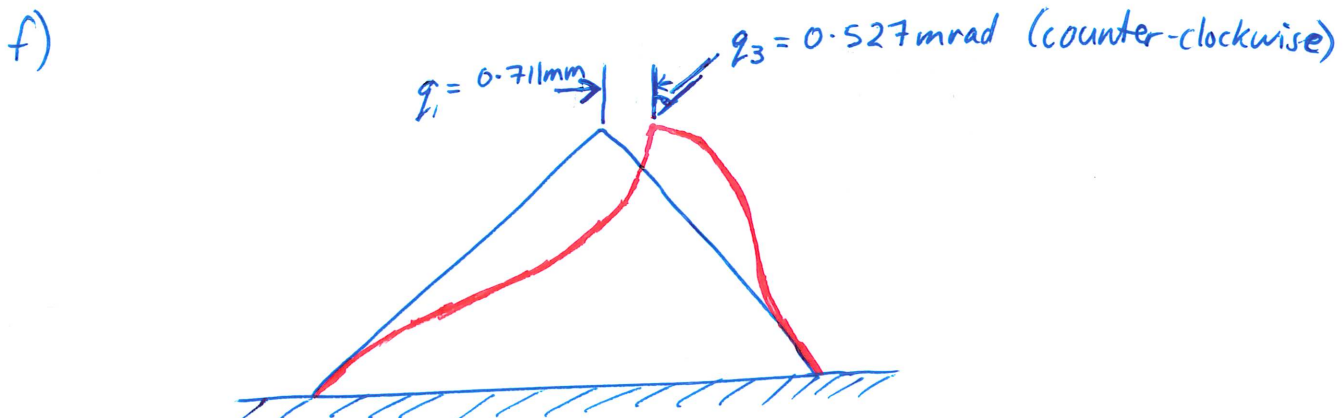
Distributed Axial Load.

$$F_{eq, DAL}' = (\Delta')^T f_{eq, DAL}' = \begin{Bmatrix} 1,800 N \\ 2,400 N \\ 0 Nm \\ 1,800 N \\ 2,400 N \\ 0 Nm \end{Bmatrix}$$

$$Q_{eq, DAL}' = A' F_{eq, DAL}' = \begin{Bmatrix} 1,800 N \\ 2,400 N \\ 0 Nm \end{Bmatrix}$$

$$Q_{total} = Q_{eq, UDL}' + Q_{eq, DAL}' = \begin{Bmatrix} 3,200 N \\ -2,400 N \\ +1,666.6 Nm \end{Bmatrix} + \begin{Bmatrix} 1,800 N \\ 2,400 N \\ 0 Nm \end{Bmatrix} = \begin{Bmatrix} 5,000 N \\ 0 N \\ +1,666.6 Nm \end{Bmatrix}$$

$$e) \quad q = \begin{Bmatrix} 0.00071097 m \\ 0 m \\ 0.00052679 rad \end{Bmatrix} = \begin{Bmatrix} 0.71097 mm \\ 0 mm \\ 0.52697 mrad \end{Bmatrix}$$



g) Left support:

$$F_{1-3}^1 + (-F_{eq,UDL,1-3}^1) + (-F_{eq,DAL,1-3}^1)$$

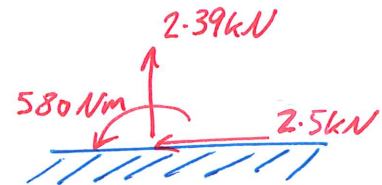
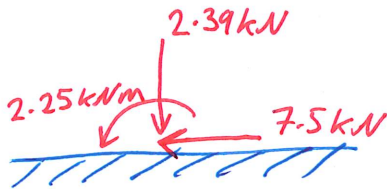
considering lost loads.

$$\begin{Bmatrix} -2,500 \text{ N} \\ -2,390.8 \text{ N} \\ 580.5 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} -3,200 \text{ N} \\ +2,400 \text{ N} \\ +1,666.6 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} -1,800 \text{ N} \\ -2,400 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

$$= \begin{Bmatrix} -7,500 \text{ N} \\ -2,390.8 \text{ N} \\ 2,247.1 \text{ Nm} \end{Bmatrix}$$

Right support:

$$F_{4-6}^2 = \begin{Bmatrix} -2,500 \text{ N} \\ +2,390.8 \text{ N} \\ 580.47 \text{ Nm} \end{Bmatrix}$$



$$f^2 = \begin{Bmatrix} 3.413 \text{ kN} \\ 0.566 \text{ kN} \\ 0.833 \text{ kNm} \\ -3.413 \text{ kN} \\ -0.566 \text{ kN} \\ 0.580 \text{ kNm} \end{Bmatrix}$$

Element [2] carries a constant shear force, so the internal bending moment varies linearly from the value at node ① to the value at node ②.

Therefore, the maximum bending moment within element [2] is 833 Nm and occurs at node ①, which is at the top of the structure.

i)

$$d^2 = \begin{Bmatrix} 0.4265795 \text{ mm} \\ 0.5687727 \text{ mm} \\ 0.5267919 \text{ mrad} \\ 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \end{Bmatrix}$$

$$\begin{aligned} \epsilon^2 &= \frac{d_4^2 - d_1^2}{L^2} = \frac{(0) - (0.4265795 \times 10^{-3})}{2.5} \\ &= -1.7063 \times 10^{-4} \epsilon \\ &= -170.6 \mu\epsilon \end{aligned}$$

as the value is negative, this axial normal strain is compressive.

- j) As the blast/wind load is the only load acting, the axial strain in element 2 will vary linearly with applied pressure.

As such, the maximum permissible loading will be:

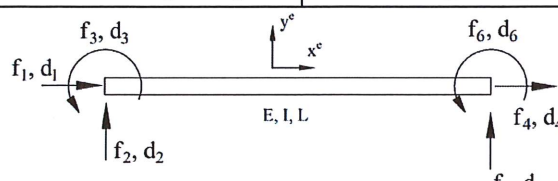
$$\frac{300}{170.6} \times 0.8 \text{ kPa} = 1.407 \text{ kPa}.$$

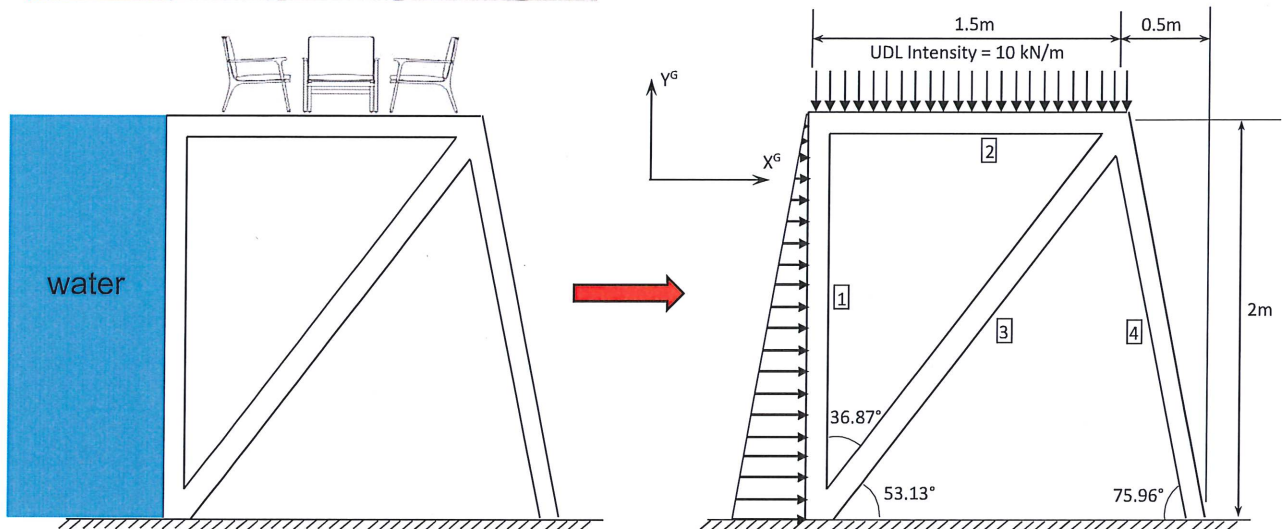
Problem 2: The diagram on the next page shows a structure that forms the side of a pool and an elevated poolside decking area. Water pressure acts on left side of the support structure and varies with the weight of water above, varying linearly from a value of 0 N/m at the top of the structure to a maximum value at the base of the structure.

The horizontal top surface supports the decking area that has a surface loading of 2kPa. The structure supports a length of decking and pool (depth into the page) of 5m, so that the magnitude of the Uniformly Distributed Load intensity is 10kN/m. The structure is made up of four frame elements and has two fixed supports at the bottom of the structure.

Assume that all elements are frame elements, made of steel, with the same cross-sectional area and second moment of area, with values given on the next page. The length of the elements differs as given on the next page, and the structure itself is considered mass-less relative to the loads it carries. **[22 Total Marks]**

- Draw a sketch of the structure showing the global degrees of freedom $\mathbf{q}$ for the model. **[2 Marks]**
- Draw the free body diagram of each element, showing the element degrees of freedom in global coordinates and the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- Write out the assembly matrices $\mathbf{A}^e$ for each element **[2 Marks]**
- Assume that the water has a density of 1,000 kg/m³, the acceleration due to gravity is 9.81 N/kg, and that the frame being modelled supports a section of pool (the distance into the page) of 5m.
Write out the element equivalent forcing terms for both elements 1 and 2 ($\mathbf{f}_{eq,LVL}^1$ and $\mathbf{f}_{eq,UDL}^2$) to represent the water pressure and the deck weight. Present the overall global forcing vector, $\mathbf{Q}$. If you cannot calculate the peak intensity of the linearly varying load, use a magnitude of $\bar{w}$ of 80 kN/m. **[4 Marks]**
- Solve for the overall deflections of the system, $\mathbf{q}$, and present your result. **[1 Mark]**
- Clearly sketch the overall deflected shape of the system.
→ In your sketch, show all the deflections and rotations that were calculated and label the magnitudes. The sketch does not need to be to scale, but each component must be labelled. **[3 Marks]**
- Provide the support reaction forces and moments at both support points, parallel and perpendicular to the supports. Provide a sketch for each case, showing the magnitude and direction of all reaction components. **[3 Marks]**
- Determine the axial normal strain in element 3 (from axial loads only, ignoring any bending effects), and indicate whether it is tensile or compressive. **[1 Mark]**
- Present the element displacement vector (in element co-ordinates) for element 3 and determine the total deflection (in the X^G and Y^G directions) of a point located at a distance of 1m (along the element) from the bottom left of element 3. The shape functions (applied to the frame element numbering system) are given below. Include your working, including presenting the values for both $u(x)$ and $v(x)$ at this location. **[4 Marks]**

List of the shape functions for a frame element.				
$\psi_1(x) = \left(1 - \frac{x}{L}\right); \psi_2(x) = \frac{x}{L}$				
$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$	$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$	$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$	$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$	
		$u(x) = \psi_1(x)d_1 + \psi_2(x)d_4$ $v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_2 \\ d_3 \\ d_5 \\ d_6 \end{pmatrix}$		



All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = E^3 = E^4 = 200 \text{ GPa}$

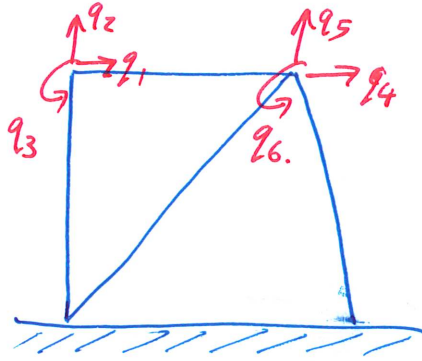
Second Moment of Area: $I^1 = I^2 = I^3 = I^4 = 3 \times 10^{-6} \text{ m}^4$

Cross-sectional Area: $A^1 = A^2 = A^3 = A^4 = 1 \times 10^{-4} \text{ m}^2$

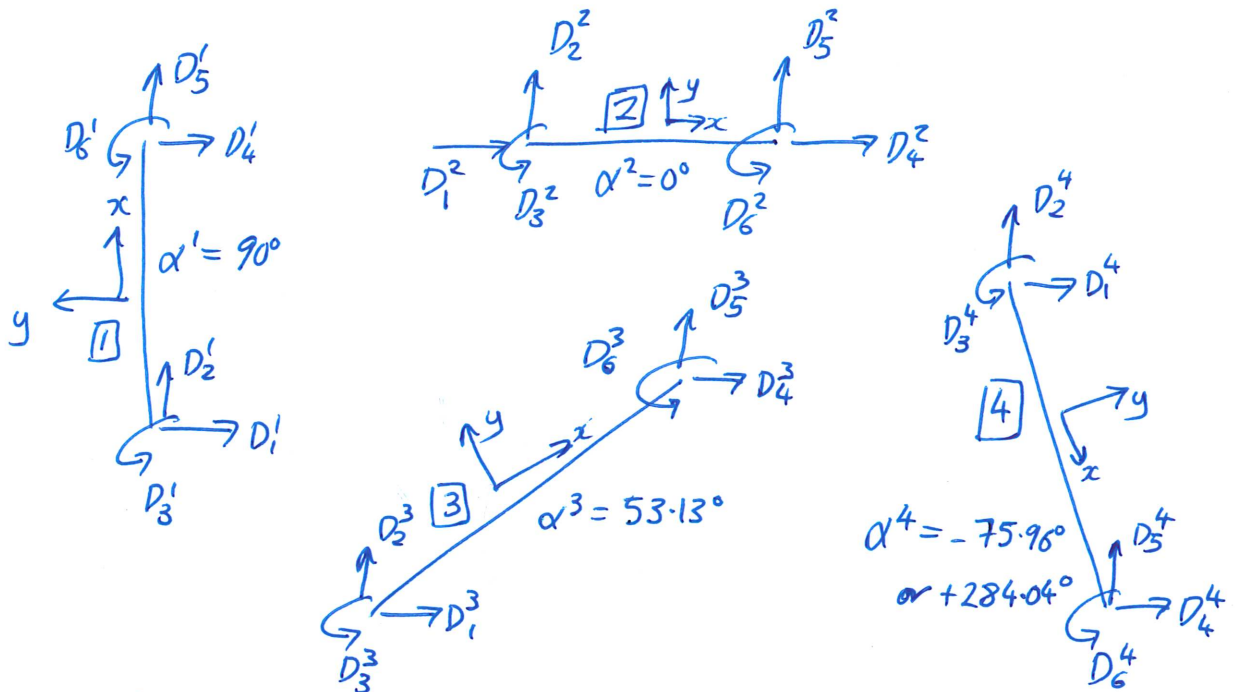
Element Lengths:

- $L^1 = 2.0\text{m}$
- $L^2 = 1.5\text{m}$
- $L^3 = 2.5\text{m}$
- $L^4 = 2.06\text{m}$

a)



b)



$$c) \quad A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = I_{6 \times 6} = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^4 = \begin{matrix} & D_1^4 & D_2^4 & D_3^4 & D_4^4 & D_5^4 & D_6^4 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$d) \quad \bar{w}_{\max, LVL} = \rho g h d = (1,000 \text{ kg/m}^3)(9.81 \text{ N/kg})(2\text{m})(5\text{m}) \\ = 98,100 \text{ N/m}.$$

Based upon my chosen orientation of element [1], with x^e for that element pointing upwards, the LVL max a maximum intensity at $x=0$ (node ①) and is zero at $x=L$ (node ②).

As such, $f_{eq, LVL}^i = \begin{Bmatrix} 0 \\ 7\bar{w}L^3/20 \\ \bar{w}(L)^3/20 \\ 0 \\ 3\bar{w}L^3/20 \\ -\bar{w}(L)^3/30 \end{Bmatrix}$ as the load from the water pressure acts in the negative y^e direction, $\bar{w} = -98,100 \text{ N/m}$.

$$f_{eq, LVL}^i = \begin{Bmatrix} 0 \\ 7(-98,100)(2)/20 \\ (-98,100)(2)^2/20 \\ 0 \\ 3(-98,100)(2)/20 \\ -(-98,100)(2)^2/30 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -68,670 \text{ N} \\ -19,620 \text{ Nm} \\ 0 \text{ N} \\ -29,430 \text{ N} \\ 13,080 \text{ Nm} \end{Bmatrix}$$

$$F_{eq, LVL}^i = (\Delta^i)^T f_{eq, LVL}^i = \begin{Bmatrix} 68,670 \text{ N} \\ 0 \text{ N} \\ -19,620 \text{ Nm} \\ 29,430 \text{ N} \\ 0 \text{ N} \\ 13,080 \text{ Nm} \end{Bmatrix}$$

$$Q_{eq, LVL}^i = A^i F_{eq, LVL}^i = \begin{Bmatrix} 29,430 \text{ N} \\ 0 \text{ N} \\ 13,080 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}.$$

d) continued.

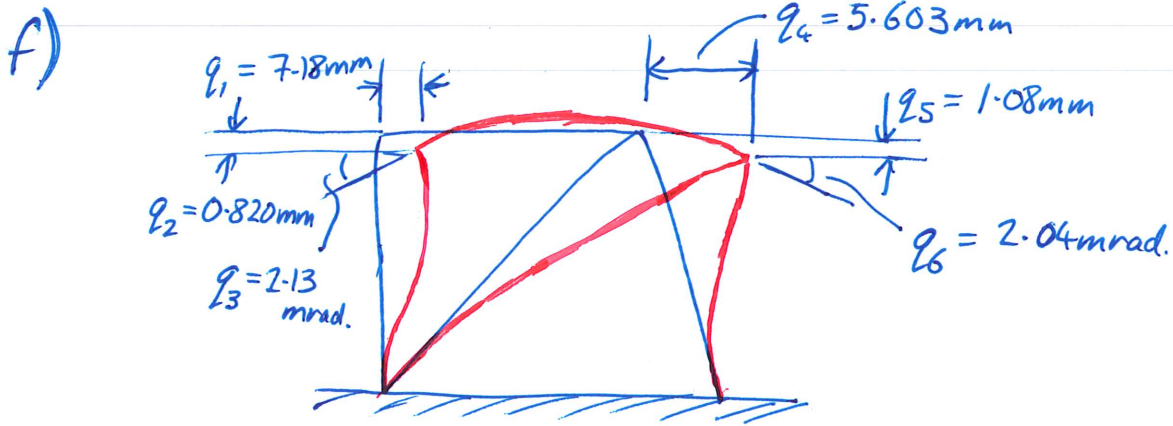
$$f_{eq,u0L}^2 = \begin{Bmatrix} 0 \\ \bar{\omega} L^2/2 \\ \bar{\omega} (L^2)^2/12 \\ 0 \\ \bar{\omega} L^2/2 \\ -\bar{\omega} (L^2)^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \\ (-10000)(1.5)/2 \\ (-10000)(1.5)^2/12 \\ 0 \\ (-10000)(1.5)/2 \\ -(-10000)(1.5)^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -7,500 \text{ N} \\ -1,875 \text{ Nm} \\ 0 \text{ N} \\ -7,500 \text{ N} \\ +1,875 \text{ Nm} \end{Bmatrix}$$

$$F_{eq,u0L}^2 = (\mathbf{1}^2)^T f_{eq,u0L}^2 = \begin{Bmatrix} 0 \text{ N} \\ -7,500 \text{ N} \\ -1,875 \text{ Nm} \\ 0 \text{ N} \\ -7,500 \text{ N} \\ +1,875 \text{ Nm} \end{Bmatrix}$$

$$Q_{eq,u0L}^2 = \mathbf{A}^2 F_{eq,u0L}^2 = \begin{Bmatrix} 0 \text{ N} \\ -7,500 \text{ N} \\ -1,875 \text{ Nm} \\ 0 \text{ N} \\ -7,500 \text{ N} \\ +1,875 \text{ Nm} \end{Bmatrix}$$

$$Q_{total} = Q_{eq,LVL} + Q_{eq,u0L}^2 = \begin{Bmatrix} 29,430 \text{ N} \\ 0 \text{ N} \\ 13,080 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ N} \\ -7,500 \text{ N} \\ -1,875 \text{ Nm} \\ 0 \text{ N} \\ -7,500 \text{ N} \\ +1,875 \text{ Nm} \end{Bmatrix} = \begin{Bmatrix} 29,430 \text{ N} \\ -7,500 \text{ N} \\ 11,205 \text{ Nm} \\ 0 \text{ N} \\ -7,500 \text{ N} \\ 1,875 \text{ Nm} \end{Bmatrix}$$

$$e) \quad q = \begin{Bmatrix} 0.00718196 \text{ m} \\ -0.00082018 \text{ m} \\ 0.002125305 \text{ rad} \\ 0.00560295 \text{ m} \\ -0.00108214 \text{ m} \\ -0.00203595 \text{ rad} \end{Bmatrix} = \begin{Bmatrix} 7.18196 \text{ mm} \\ -0.82018 \text{ mm} \\ 2.12530 \text{ mm} \\ 5.60295 \text{ mm} \\ -1.08214 \text{ mm} \\ -2.03595 \text{ mm} \end{Bmatrix}$$



g) Left support:

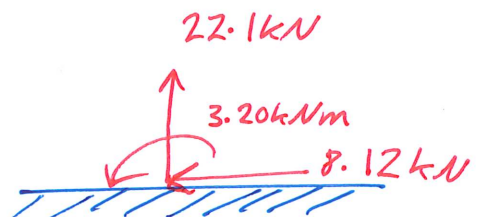
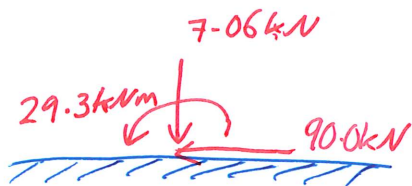
$$F'_{1-3} + F^3_{1-3} + (-F'_{q, LVK, 1-3})$$

$$\begin{pmatrix} -8.3765 \text{ kN} \\ 8.2018 \text{ kN} \\ 7.7389 \text{ kNm} \end{pmatrix} + \begin{pmatrix} -12.9346 \text{ kN} \\ -15.2596 \text{ kN} \\ 1.9786 \text{ kNm} \end{pmatrix} + \begin{pmatrix} -68.670 \text{ kN} \\ 0 \text{ kN} \\ +19.62 \text{ kNm} \end{pmatrix}$$

$$= \begin{pmatrix} -89.981 \text{ kN} \\ -7.0578 \text{ kN} \\ 29.3475 \text{ kNm} \end{pmatrix}$$

Right support:

$$F^4_{4-6} = \begin{pmatrix} -8.1188 \text{ kN} \\ 22.0577 \text{ kN} \\ 3.1969 \text{ kNm} \end{pmatrix}$$



h)

$$d^3 = \begin{pmatrix} 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \\ 2.4961 \text{ mm} \\ -5.1316 \text{ mm} \\ -2.0359 \text{ mrad} \end{pmatrix}$$

$$\epsilon^3 = \frac{d^3_4 - d^3_1}{L^3} = \frac{(2.4961 \times 10^{-3}) - (0)}{2.5}$$

$$\epsilon^3 = 9.9844 \times 10^{-4} \epsilon$$

$$\epsilon^3 = 998.4 \mu \epsilon$$

this strain is positive, which means that it is tensile.

$$j) \quad d^3 = \begin{Bmatrix} 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \\ 2.4961 \text{ mm} \\ -5.1316 \text{ mm} \\ -2.0359 \text{ mrad} \end{Bmatrix}$$

$$u(x) = \psi_1(x) d_1 + \psi_2(x) d_4$$

$$v(x) = \mathcal{N}_1(x) d_2 + \mathcal{N}_2(x) d_3 + \mathcal{N}_3(x) d_5 + \mathcal{N}_4(x) d_6$$

$$\psi_2(x=1\text{m}) = \frac{1.0}{2.5} = 0.4$$

$$\mathcal{N}_3(x=1\text{m}) = \frac{3(1)^2}{(2.5)^2} - \frac{2(1)^3}{(2.5)^3} = 0.352$$

$$\mathcal{N}_4(x=1\text{m}) = \frac{(1)^3}{(2.5)^2} - \frac{(1)^2}{(2.5)} = -0.24$$

$$u(x) = \psi_1(x)(0) + (0.4)(2.4961) = 0.9984 \text{ mm}$$

$$v(x) = \mathcal{N}_1(x)(0) + \mathcal{N}_2(x)(0) + (0.352)(-5.1316) + (-0.24)(-2.0359)$$

$$v(x) = -1.318 \text{ mm}$$

$$\delta X_G = u(x) \cos(\alpha) - v(x) \sin(\alpha)$$

$$= (0.9984) \cos(53.13^\circ) - (-1.318) (\sin(53.13^\circ))$$

$$= 1.653 \text{ mm}$$

$$\delta Y_G = u(x) \sin(\alpha) + v(x) \cos(\alpha)$$

$$= (0.9984) \sin(53.13^\circ) + (-1.318) (\cos(53.13^\circ))$$

$$= 0.007917 \text{ mm}$$